Team Name: DermAI

Project Title: Quantifying Explanation Faithfulness and Localization in CNNs vs. Vision Transformers for Skin Lesion Classification

Group Members:

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Project Summary:

Deep learning models now match dermatologist-level accuracy at skin-lesion classification (Esteva et al., 2017), yet clinical adoption remains low because clinicians cannot verify which visual features drive a model's decision. This lack of interpretability is especially costly given that early detection significantly improves survival rates (5-year melanoma survival exceeds 95% if caught early but drops below 30% once metastatic), making reliable and explainable AI screening tools clinically significant. This project develops and compares two architecturally distinct classifiers, a CNN (EfficientNet-B0) and a Vision Transformer (ViT-B/16), on multi-class dermoscopic image classification. CNNs exploit local spatial structure through convolution, while ViTs capture global dependencies between distant image patches via attention, making the two architectures complementary and well-suited to this problem. Beyond classification accuracy, we investigate explanation quality using Grad-CAM for the CNN and Attention Rollout for the ViT, evaluated along two axes: faithfulness, via deletion and insertion tests that measure whether highlighted regions actually drive the prediction, and localization, via Intersection-over-Union and the pointing game against ground-truth lesion masks.

Approach:

We will use pretrained models obtained from Hugging Face with the same preprocessing pipeline, augmentation strategy (random flips, rotation, color jitter, and cropping), weighted cross-entropy loss, and evaluation protocol. Training follows a two-stage strategy: the backbone is first frozen while only the classification head is trained, then the full model is fine-tuned end-to-end at a reduced learning rate, which is especially important for ViT to avoid destabilizing pretrained attention weights on small datasets. Attribution maps from both models will be compared qualitatively against clinically meaningful lesion features such as border irregularity and color heterogeneity. The core training loop, evaluation pipeline, and attribution map implementation will be written by the team, using torchvision for model definitions and augmentation utilities. Stretch goals include replacing weighted loss with focal loss and domain-specific augmentation such as random erasing to simulate hair artifacts and CutMix for minority class augmentation.

- EfficientNet-B0 from https://huggingface.co/google/efficientnet-b0
- ViT-B/16 from https://huggingface.co/google/vit-base-patch16-224

Resources / Related Work & Papers

1. Kuntal & Bhat (2025), "Bridging CNNs and Transformers for Skin Cancer Detection," IEEE ISCON. Our direct baseline: CNN vs. Transformer on HAM10000, accuracy and efficiency only, no explainability.
2. Tschandl et al. (2018), "The HAM10000 Dataset," Scientific Data. Dataset reference and source of lesion segmentation masks used as localization ground truth.
3. Selvaraju et al. (2017), "Grad-CAM," ICCV. Our CNN explanation method.
4. Abnar & Zuidema (2020), "Quantifying Attention Flow in Transformers," ACL. Defines attention rollout, our ViT explanation method.
5. Petsiuk et al. (2018), "RISE," BMVC. Origin of the deletion/insertion faithfulness metrics.
6. Dosovitskiy et al. (2021), "An Image is Worth 16x16 Words," ICLR. The Vision Transformer architecture.
7. Tan & Le (2019), "EfficientNet," ICML. The CNN backbone.
8. Esteva et al. (2017), "Dermatologist-level classification of skin cancer," Nature. Foundational clinical motivation.
9. Kozachok & Seregin (2026), "Clinical Validation of the Melanoscope AI," arXiv. Closest related work: CNN vs. Transformer heatmap IoU against expert annotations on ~180 cases, no faithfulness metrics, framed as product validation. Read to differentiate our contribution.

Datasets:

1. Dataset overview: **HAM1000**, containing 10,015 dermoscopic images across 7 diagnostic categories. Available via Harvard Dataverse/Kaggle.
2. Ground truth masks: Binary lesion segmentation masks (DOI: 10.7910/DVN/DBW86T) serve as absolute ground truth for localization metrics.
3. Data splitting: Enforcing a lesion-grouped stratified 80/10/10 split grouped by lesion_id.
4. Key characteristics: Features severe class imbalance (~67% nevi vs. ~1% vascular lesions), utilize RGB images resized to 224x224.

Evaluation Plan

Models will be evaluated under identical conditions across 3 random seeds:

1. Classification metrics: Per-class AUC-ROC, macro F1, balanced accuracy, and failure-case overlap to analyze the complementary strengths of CNN vs. ViT representations.
2. Explainability (faithfulness): Deletion and insertion AUC to mathematically verify if the heatmap's highlighted pixels actually drive the model's prediction probability.

3. Explainability (localization): Intersection-over-Union (IoU) and pointing gate hit-rate evaluated against ground truth segmentation masks to verify if the heatmap lands accurately on the lesion.
4. Qualitative analysis: A curated panel of clear successes and instructional failures to contextualize the quantitative findings.

Risks and Scope

Three primary risks are identified:

1. Class imbalance may inflate overall accuracy while harming minority class recall, mitigated by weighted cross-entropy loss and per-class F1 monitoring.
2. ViT-B/16's limited data efficiency on a dataset far smaller than its pretraining corpus, mitigated by ImageNet initialization and two-stage fine-tuning, with any remaining performance gap treated as an informative finding.
3. Compute constraints on PACE with a fallback to EfficientNet-only training and deeper ablation if needed.

Required scope includes training and evaluating both EfficientNet-B0 and ViT-B/16 on HAM10000, Grad-CAM and Attention Rollout structured evaluation, quantitative comparison via AUC-ROC, macro F1, and balanced accuracy (mean ± std, 3 seeds), a minimum ablation study on loss function and fine-tuning strategy, and a documented iteration log. Stretch goals include a focal loss ablation, domain-specific dermoscopy augmentation, and an EfficientNet and ViT ensemble.

Ethical Implications

Automated skin lesion classification carries meaningful ethical risk. HAM10000 was collected primarily from European institutions, so underrepresented skin tones are less prevalent in training data, and a model underperforming on these groups could contribute to diagnostic disparities if deployed clinically. We will acknowledge this limitation explicitly and frame this project strictly as a benchmark comparison and explainability study, not a clinical tool. The Grad-CAM and Attention Rollout components address the concern that opaque predictions are unsafe in clinical contexts, and our explainability analysis is intended as a tool for detecting failure modes such as automation bias, not as a guarantee of model safety.